



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO M/S S.S AGENCY

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

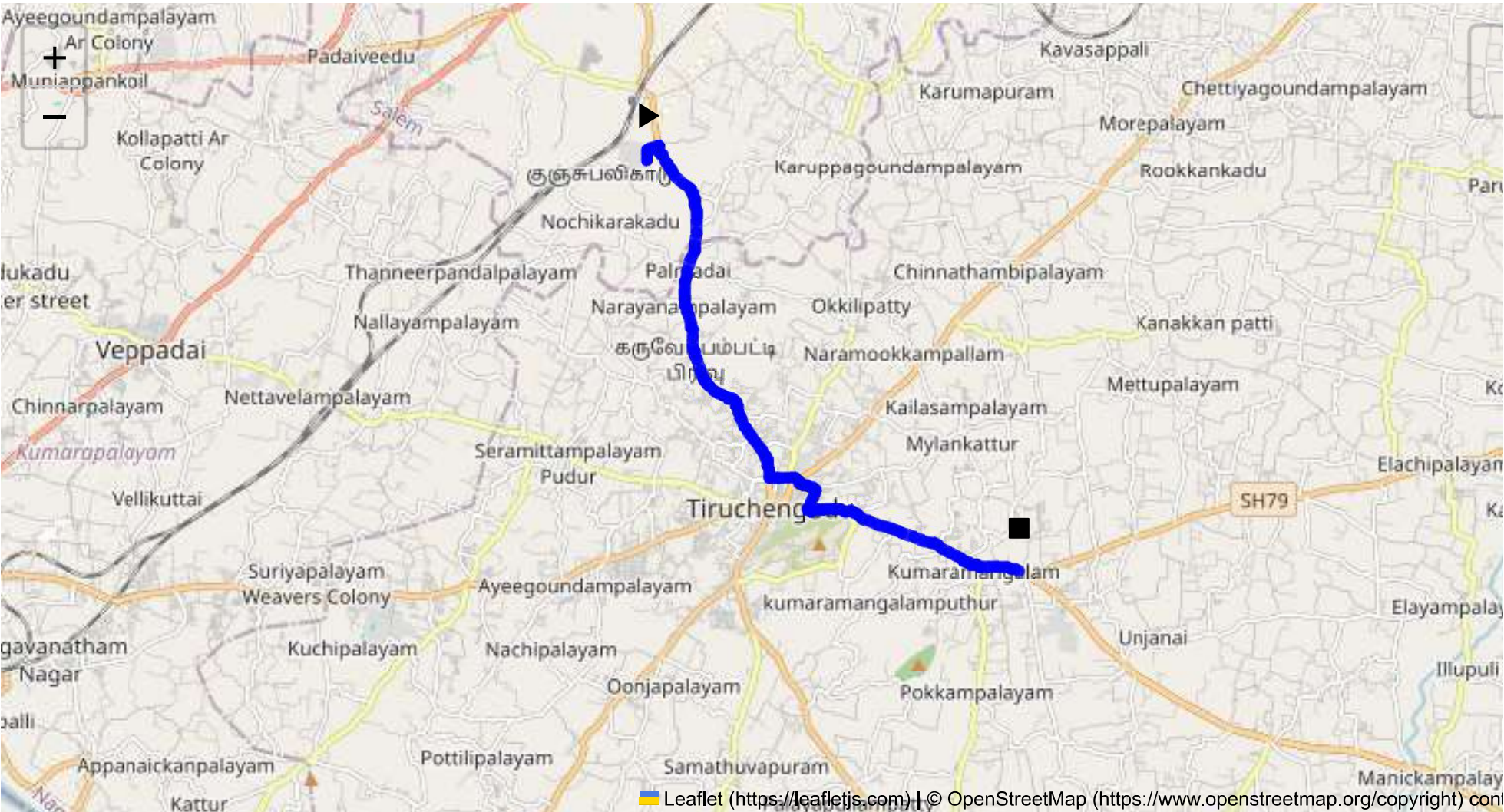
## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 13.05 km**  
**Estimated Duration: 0.4 hours**  
**Adjusted Duration (Heavy Vehicle): 0.5 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.36762, 77.937744)**



Welcome to the Journey Risk Management Study

Route Overview

The route spans approximately 13.05 kilometers from CVQF+23W, Sangagiri to 9W9Q+44J, in Nadar Theru, along the Trichy - Namakkal - Tiruchengode Road in Tamil Nadu. This fairly short stretch primarily involves local roads with some state highway segments.



# Weather Conditions

- **Typical Weather:** The region experiences a tropical climate with high temperatures throughout the year. The monsoon season, from June to September, brings heavy rain which can lead to road flooding.
- **Weather-Related Hazards:** Heavy rains could reduce visibility and make road surfaces slippery. Potential flooding might result in road closures or diversions.

# Traffic Patterns

- **Peak Hours:** Morning (8:00-10:00 AM) and evening (5:00-7:00 PM) are peak congestion periods, especially at intersections and market areas.
- **Congestion-Prone Areas:** Points near schools, local markets, and junctions connecting to larger highways could see significant slow-downs.

# Road Quality and Infrastructure

- **Assessment:** The roads are generally in good condition but may have occasional potholes especially after the monsoon. The highway sections are well-maintained, while some local roads might lack proper signage and lighting.
- **Infrastructure:** Adequate signage, though not necessarily up-to-date. Limited roadside amenities for truck drivers like rest stops or fuel stations.

# Alternative Routes for Emergencies

- **Suggestion:** In case of blockage, take National Highway 544 till Namakkal, then local links which can bypass any specific problem areas en route.
- **Backup:** Local roads parallel to the main route offer potential detours, though may not be suitable for heavy vehicles.

# Regulations for Hazardous Material Transport

- **Local Regulations:** Adherence to specific time-bound movements for hazardous materials to reduce public exposure (typically restricted during peak hours).
- **Permits and Documentation:** Ensure all required permits are up to date and comply with the Central Motor Vehicles Rules applicable in India.

# Historical Incidents

- **Overview:** No major recent incidents specifically linked to heavy vehicles or hazardous materials in the past few years along this route.
- **Past Issues:** Occasional reports of minor accidents primarily due to human error or adverse weather conditions.

## Environmental Considerations

- **Sensitive Areas:** Care near agricultural zones to prevent contamination from hazardous material spills.
- **Conservation Zones:** Limited wildlife crossings or protected zones; however, vigilance is recommended to avoid any ecological damage.

## Communication Coverage

- **Coverage Analysis:** The route has fairly consistent mobile network coverage with some potential drop zones in more rural or less populated segments.
- **Dead Zones:** Primarily in regions with geographical impediments or sparse population.

## Emergency Response Times

- **Response Estimates:** Emergency response, including fire and medical services, can take 20-30 minutes to reach rural segments. Urban areas closer to Sangagiri and Tiruchengode are likely to have quicker response times (~15 minutes).

## Overall Risk Assessment Summary

Overall, the route is moderately safe for transporting hazardous materials. Key risks involve weather-related challenges, congestion during peak hours, and potential communication dead zones. Recommended strategies for mitigating risks include avoiding travel during peak weather seasons, maintaining regular communication checks, adhering strictly to local transport regulations, and mapping out alternative routes in advance.

It is advisable to conduct detailed checks on vehicle condition, carry communication equipment, and ensure drivers are trained in emergency protocols specific to hazardous materials.

### Risk Assessment - Turns

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 0 | Roundabout | High       | 11.38395, 77.89480 | 15 KM/Hr    |
| 1 | Turn       | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 2 | Turn       | Medium     | 11.43956, 77.87340 | 30 KM/Hr    |
| 3 | Turn       | Medium     | 11.43971, 77.87348 | 30 KM/Hr    |
| 4 | Blind Spot | Blind Spot | 11.44029, 77.87544 | 10 KM/Hr    |
| 5 | Turn       | High       | 11.38346, 77.89975 | 15 KM/Hr    |
| 6 | Turn       | Medium     | 11.38190, 77.90254 | 30 KM/Hr    |

|   | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|---|------------|------------|--------------------|-------------|
| 7 | Turn       | Medium     | 11.38186, 77.90258 | 30 KM/Hr    |
| 8 | Turn       | Medium     | 11.38182, 77.90259 | 30 KM/Hr    |
| 9 | Blind Spot | Blind Spot | 11.37841, 77.90140 | 10 KM/Hr    |

## Emergency Locations

|   | type     | name                             | coordinates            | speed_limit | risk_level |
|---|----------|----------------------------------|------------------------|-------------|------------|
| 0 | hospital | Tiruchengode, Goverment Hospital | 11.3903328, 77.8920627 | 30 km/h     | Medium     |
| 1 | hospital | SPM Medical Centre,Tiruchengode  | 11.3881331, 77.8931963 | 30 km/h     | Medium     |
| 5 | hospital | Jayaa Hospital                   | 11.3848223, 77.9021886 | 30 km/h     | Medium     |

## Crowded Spots

|   | type        | name                                | coordinates            | speed_limit | risk_level |
|---|-------------|-------------------------------------|------------------------|-------------|------------|
| 2 | school      | அரசு ஆண்கள் மேல்நிலைப் பள்ளி        | 11.3850439, 77.8948279 | 30 km/h     | Medium     |
| 3 | school      | அரசு பெண்கள் மேல்நிலைப் பள்ளி       | 11.3844247, 77.8949053 | 30 km/h     | Medium     |
| 4 | marketplace | திருச்செங்கோடு தினசரி காய்கறி சந்தை | 11.3833608, 77.8970145 | 30 km/h     | Medium     |

## Route Photos of Risky Spots